

Scaling Limits of Markov chains on Integer Compositions

Quan Shi

AMSS, Chinese Academy of Sciences

quan.shi@amss.ac.cn

Conference Discrete Random Structures II
Będlewo, May 22, 2026

Overview

1. Scaling limits of Chinese Restaurant Processes

Joint work with



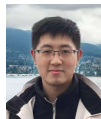
Matthias Winkel
(Oxford)

2. Construct the limiting process via a Brownian motion

Joint work with



Elie Aïdékon
(Shanghai)

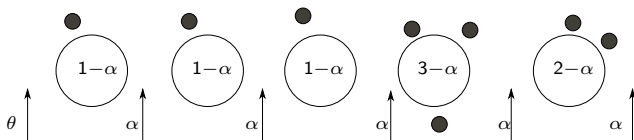


Chengshi Wang
(Shanghai)

1. Scaling limits of Chinese Restaurant Processes

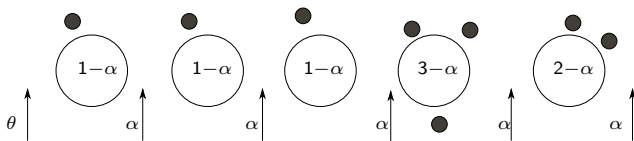
Ordered Chinese Restaurant Processes

- ▶ $\alpha \in (0, 1)$, $\theta \geq 0$
- ▶ Tables are **ordered** in a line.
- ▶ When a new customer arrives:
choose an occupied table or **start a new table**, proportional to the **weight**:
- ▶ ▶ Each table of $m \in \mathbb{N}$ customers: weight $m - \alpha$
- ▶ ▶ A new table at the leftmost position: weight θ
- ▶ ▶ A new table between each pair of two neighbouring occupied tables or at the rightmost position: weight α ;



Ordered Chinese Restaurant Processes

- ▶ $\alpha \in (0, 1)$, $\theta \geq 0$
- ▶ Tables are **ordered** in a line.
- ▶ When a new customer arrives:
choose an occupied table or **start a new table**, proportional to the **weight**:
- ▶ ▶ Each table of $m \in \mathbb{N}$ customers: weight $m - \alpha$
- ▶ ▶ A new table at the leftmost position: weight θ
- ▶ ▶ A new table between each pair of two neighbouring occupied tables or at the rightmost position: weight α ;



- ▶ The attribution of $n \in \mathbb{N}$ customers from left to right:
 $\sigma_n = (\sigma_1, \dots, \sigma_k)$ with $\sigma_1 + \dots + \sigma_k = n$
- ▶ σ_n is a **composition of integer $n \in \mathbb{N}$**

A related tree-growth model [Pitman-Winkel 2009]

- ▶ Add leaves one by one:
weight α on inner edge; weight $1 - \alpha$ on leaf edge; weight θ on edge of 0.
- ▶ Order Chinese restaurant process: sizes of the subtrees (number of leaves) along the path $0 \rightarrow \text{root}$.
- ▶ $\alpha = \theta = \frac{1}{2}$: Rémi's algorithm
- ▶ $\theta = 1 - \alpha$: Ford's alpha model

A composition structure

- ▶ $(\sigma_n, n \geq 1)$ is a **regenerative composition structure** [Gnedin 1997, Gnedin-Pitman 2005]
- ▶ Convergence to **Poisson-Dirichlet interval partition** of $[0, 1]$:

$$\frac{1}{n}\sigma_n \rightarrow \text{PDIP}_{\alpha,\theta}, \quad \text{a.s.}$$

$\text{PDIP}_{\alpha,\theta} = \text{closure of } \{1 - \exp(-Y_t), t \geq 0\}$ for a **subordinator** Y .

A composition structure

- ▶ $(\sigma_n, n \geq 1)$ is a **regenerative composition structure** [Gnedin 1997, Gnedin-Pitman 2005]
- ▶ Convergence to **Poisson-Dirichlet interval partition** of $[0, 1]$:

$$\frac{1}{n}\sigma_n \rightarrow \text{PDIP}_{\alpha,\theta}, \quad \text{a.s.}$$

$\text{PDIP}_{\alpha,\theta} = \text{closure of } \{1 - \exp(-Y_t), t \geq 0\}$ for a **subordinator** Y .

- ▶ An **interval partition** of the interval $[0, L]$:
 $\beta = \{(a_i, b_i) \subset (0, L) : i \geq 1\}$ a collection of disjoint open intervals, with **total mass** (sum of lengths) $\|\beta\| := \sum_{i \geq 1} (b_i - a_i) = L$.
- ▶ Under Hausdorff metric: a polish space
- ▶ **Composition** $(1, 1, 1, 3, 2) \Leftrightarrow$ **Interval Partition** 

A composition structure

- ▶ $(\sigma_n, n \geq 1)$ is a **regenerative composition structure** [Gnedin 1997, Gnedin-Pitman 2005]
- ▶ Convergence to **Poisson-Dirichlet interval partition** of $[0, 1]$:

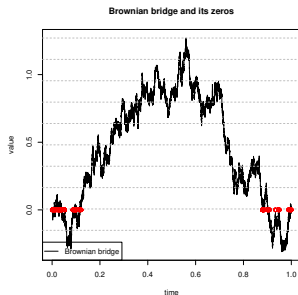
$$\frac{1}{n}\sigma_n \rightarrow \text{PDIP}_{\alpha,\theta}, \quad \text{a.s.}$$

$\text{PDIP}_{\alpha,\theta} = \text{closure of } \{1 - \exp(-Y_t), t \geq 0\}$ for a **subordinator** Y .

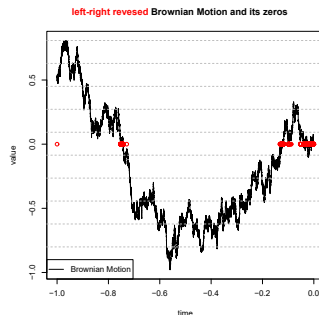
- ▶ An **interval partition** of the interval $[0, L]$:
 $\beta = \{(a_i, b_i) \subset (0, L) : i \geq 1\}$ a collection of disjoint open intervals, with **total mass** (sum of lengths) $\|\beta\| := \sum_{i \geq 1} (b_i - a_i) = L$.
- ▶ Under Hausdorff metric: a polish space
- ▶ **Composition** $(1, 1, 1, 3, 2) \Leftrightarrow$ **Interval Partition** 
- ▶ $\text{PDIP}_{\alpha,\theta}$ project to Kingman's simplex: Poisson-Dirichlet distribution (α, θ)

Examples of PDIPs

$\text{PDIP}(1/2, 1/2)$: zero points of a Brownian bridge on $[0, 1]$ from zero to zero.

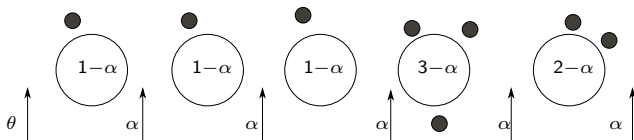


$\text{PDIP}(0, 1/2)$: zero points of a Brownian motion on $[0, 1]$; the left-right reversal.



Ordered Up-Down Chain

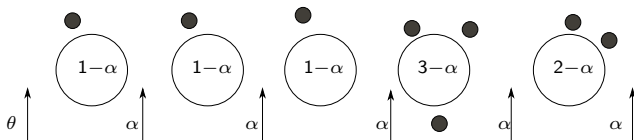
- ▶ $\alpha \in (0, 1)$, $\theta \geq 0$
- ▶ Tables are ordered in a line.
- ▶ At each step, one arrives (up) and then one leaves (down)
- ▶ Up: the same (α, θ) arrival rule



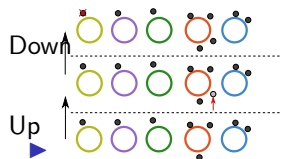
- ▶ Down: Each customer leaves at rate 1.

Ordered Up-Down Chain

- ▶ $\alpha \in (0, 1)$, $\theta \geq 0$
- ▶ Tables are ordered in a line.
- ▶ At each step, one arrives (**up**) and then one leaves (**down**)
- ▶ **Up**: the same (α, θ) arrival rule



- ▶ **Down**: Each customer leaves at rate 1.



$(1, 1, 4, 2)$



$(1, 1, 1, 4, 2)$



$(1, 1, 1, 3, 2)$



- For each $n \in \mathbb{N}$, the up-down chain $(C^n(k), k \geq 0)$ has stationary distribution σ_n given by the Chinese restaurant process.

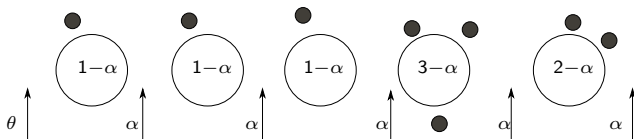
Theorem (Rivera-Lopez&Rizzolo, 2022)

$$\left(\frac{1}{n} C^n(\lfloor n^2 t \rfloor), t \geq 0\right) \rightarrow \left(\bar{\beta}(t), t \geq 0\right) \quad \text{in distribution}$$

- the limit is a continuous Markov process on the space of unit interval partitions, with **stationary distribution** $\text{PDIP}_{\alpha, \theta}$.
- Related results for **integer partition** chains:
Diffusion on the Kingman simplex (Borodin, Gnedenko, Olshanski, Petrov ...)

Poissonization

- ▶ Tables are ordered in a line.
- ▶ $\alpha \in (0, 1)$, $\theta \geq 0$. Construct a **continuous-time** Markov chain.
- ▶ **Arriving (up-step)**:
 - ▶ For each occupied table, say there are $m \in \mathbb{N}$ customers, a new customer comes to join this table at **rate** $m - \alpha$
 - ▶ At **rate** θ , a new customer enters to start a new table at the leftmost position.
 - ▶ Between each pair of two neighbouring occupied tables or at the rightmost position, a new customer enters and begins a new table there at **rate** α ;



- ▶ **Leaving (down-step)**: Each customer leaves at **rate** 1.

Theorem (Forman-Rizzolo-Pal-Winkel, 2021,S.–Winkel 2026)

$$\left(\frac{C(2nt)}{n}, t \geq 0 \right) \xrightarrow{n \rightarrow \infty} (\beta(t), t \geq 0) \quad \text{in distribution.}$$

The **scaling limit** $(\beta(t), t \geq 0)$ is an **interval-partition-valued diffusion process** $\text{IPE}(\alpha, \theta)$.

► Applications:

Fleming-Viot processes conjectured by Feng-Sun

[Forman-Rizzolo-S.-Winkel, 2022]

Continuum-tree-valued process conjectured by Aldous

[Forman-Rizzolo-Pal-Winkel, 2023+]

A simulation of an Interval-Partition Evolution:

http://www.stats.ox.ac.uk/~winkel/5_sim_skewer.gif

Construction of IPE

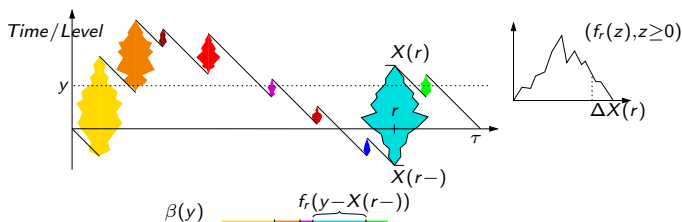
“Scaffolding and Spindles” from [Forman et. al. EJP2020]

- ▶ “Scaffolding” X : spectrally positive $(1 + \alpha)$ -stable Lévy process, with Laplace exponent $\psi(\lambda) = \frac{\lambda^{1+\alpha}}{2^\alpha \Gamma(1+\alpha)}$

- ▶ “Spindles”: Mark each jump $(t, \Delta X_t)$ of X , by a $\text{BESQ}(-2\alpha)$ -excursion of length ΔX_t :

$\mathbf{N} = \sum_{i \in \mathbb{N}} \delta_{(t_i, f_i)}$: Poisson point process of $\text{BESQ}(-2\alpha)$ -excursions, such that

$$\sum_{\Delta X(t) > 0} \delta_{(t, \Delta X(t))} = \sum_{i \in \mathbb{N}} \delta_{(t_i, \zeta(f_i))}.$$



De-Poissonized process

- ▶ Introduce a Lamperti-type **time-change**

$$\tau(u) := \inf \left\{ y \geq 0: \int_0^y \|\beta(z)\| dz > u \right\}, \quad u \geq 0.$$

- ▶ (Forman-Rizzolo-S.-Winkel 2023) The **de-Poissonized process**

$$\bar{\beta}(u) := \|\beta(\tau(u))\|^{-1} \beta(\tau(u)), \quad u \geq 0$$

is the same as the limit process of [Rivera-Lopez–Rizzolo 2022].

De-Poissonized process

- ▶ Introduce a Lamperti-type **time-change**

$$\tau(u) := \inf \left\{ y \geq 0 : \int_0^y \|\beta(z)\| dz > u \right\}, \quad u \geq 0.$$

- ▶ (Forman-Rizzolo-S.-Winkel 2023) The **de-Poissonized process**

$$\bar{\beta}(u) := \|\beta(\tau(u))\|^{-1} \beta(\tau(u)), \quad u \geq 0$$

is the same as the limit process of [Rivera-Lopez–Rizzolo 2022].

- ▶ **Conjecture** [Forman-Rizzolo-Pal-Winkel, 2021]:
The process $\bar{\beta}$ is independent of the total mass process $\|\beta\|$.

De-Poissonized process

- ▶ Introduce a Lamperti-type **time-change**

$$\tau(u) := \inf \left\{ y \geq 0 : \int_0^y \|\beta(z)\| dz > u \right\}, \quad u \geq 0.$$

- ▶ (Forman-Rizzolo-S.-Winkel 2023) The **de-Poissonized process**

$$\bar{\beta}(u) := \|\beta(\tau(u))\|^{-1} \beta(\tau(u)), \quad u \geq 0$$

is the same as the limit process of [Rivera-Lopez–Rizzolo 2022].

- ▶ **Conjecture** [Forman-Rizzolo-Pal-Winkel, 2021]:
The process $\bar{\beta}$ is independent of the total mass process $\|\beta\|$.
- ▶ Solved by [Aïdékon-S.Wang 2026+], but by using a new construction.

2. Construction via a Brownian motion

A space-filling curve exploring a white noise

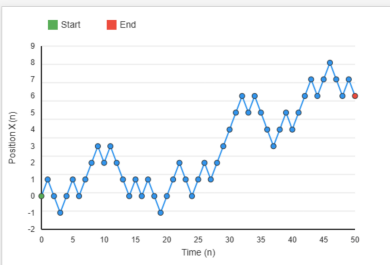
- ▶ **local time** of a Brownian motion B at level $y \in \mathbb{R}$ until time $t \geq 0$:

$$L(t, y) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \int_0^t \mathbb{1}_{\{B_s \in [y, y+\epsilon]\}} ds.$$

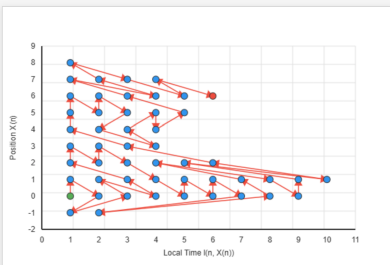
- ▶ $\rho_t := (L(t, B_t), B_t), t \geq 0$: **A space-filling curve** on $\mathbb{R}_+ \times \mathbb{R}$, with continuous path, and does not cross itself.
- ▶ **A white noise** on $\mathbb{R}_+ \times \mathbb{R}$: $\forall g \in L^2(\mathbb{R}_+ \times \mathbb{R}, \text{Leb})$

$$W(g) := \int_0^\infty g(\rho_t) \text{sgn}(B_t) dB_t$$

Random Walk on $\mathbb{Z}$



Path: $(l(n, X(n)), X(n))$



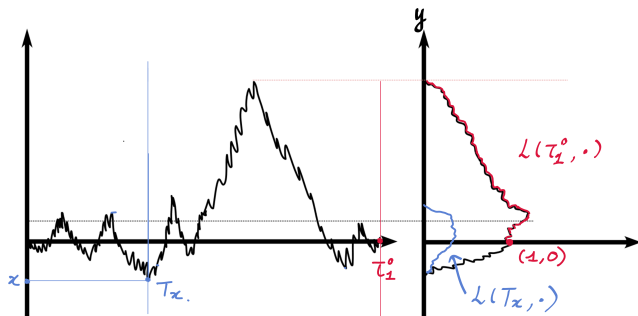
Ray–Knight Theorems

Squared Bessel process Z with dimension $\delta \in \mathbb{R}$, $\text{BESQ}(\delta)$:

$$dZ_t = 2\sqrt{|Z_t|}d\beta_t + \delta dt.$$

Theorem (Ray–Knight)

1. $x \leq 0$. T_x : the first hitting time of B at x
The process $(L(T_x, x + y), y \in [0, |x|])$: $\text{BESQ}(2)$ starting from 0.
2. $z \geq 0$. $\tau_z^0 = \inf\{t \geq 0 : L(t, 0) > z\}$
The process $(L(\tau_z^0, y), y \geq 0)$: $\text{BESQ}(0)$ starting from z .



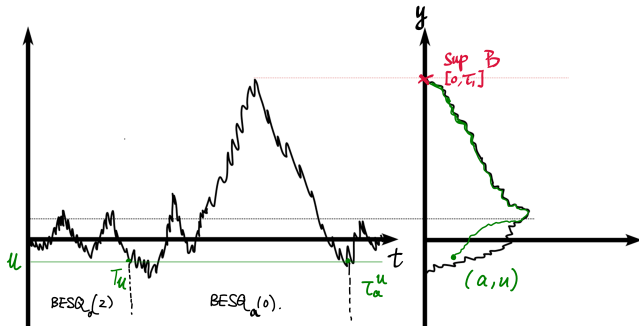
Ray-Knight Theorems (conti.)

Corollary

$a \geq 0, u \in \mathbb{R}$. $\tau_a^u = \inf\{t \geq 0: L(t, u) > a\}$

The process $(L(\tau_a^u, u + y), y \geq 0)$:

BESQ(2) when $u + y \leq 0$; BESQ(0) when $u + y \geq 0$.



Ray–Knight Theorems for PRBM

[LeGall–Yor1986, Carmona–Petit–Yor1994]

For perturbed reflected Brownian motion (PRBM) ξ : for $\mu > 0$,

$$\xi_t = |B_t| - \mu L(t, 0)$$

- ▶ $L_\xi(t, u)$: local time of ξ .
 $a \geq 0, u \in \mathbb{R}$. $\tau_a^u = \inf\{t \geq 0: L_\xi(t, u) > a\}$
- ▶ The process $(L_\xi(\tau_a^u, y), y \geq u)$:
BESQ(δ) when $y \leq 0$; BESQ(0) when $y \geq 0$: denoted by
BESQ($\delta \mid_0 0$)
- ▶ $\delta = 2/\mu > 0$ ($\delta = 2$ for BM)

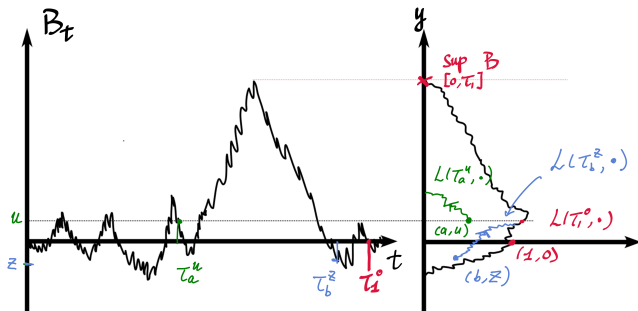
An infinite-dimensional representation

Theorem (Aidékon-Hu-Shi, 2023)

$\tau_a^u = \inf\{t \geq 0 : L_\xi(t, u) > a\}$. Almost surely for all $a \geq 0, u \in \mathbb{R}$: The process $(L_\xi(\tau_a^u, y), y \geq u)$ is strong solution of the SDE

$$L_\xi(\tau_a^u, y) = a + 2 \int_u^y W([0, L_\xi(\tau_a^u, s)], ds) + (\delta \mathbb{1}_{\{y \leq 0\}} + 0 \mathbb{1}_{\{y \geq 0\}})(y - u),$$

The family $\{L_\xi(\tau_a^u, y)_{y \geq u}, (a, u) \in \mathbb{R}_+ \times \mathbb{R}\}$: BESQ(δ | 0) flows



- ▶ Stochastic flows connected to continuous-state branching processes and coalescent:
[Pitman-Yor 82] [Bertoin-Le Gall 00, 03, 05], [Dawson-Li,12]
[Foucart-Ma-Mallein 19] and others ...

Applications of [Aïdékon-Hu-Shi, 23]:

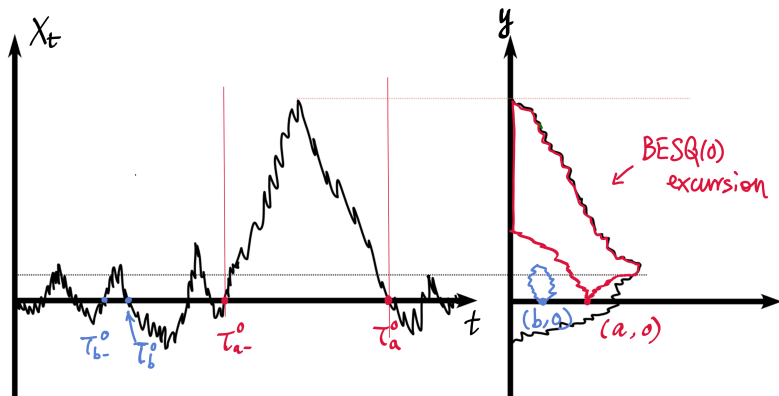
- ▶ [Aïdékon-Hu-Shi, 26+]:
Brownian loop soups [Lupu-Sabot-Tarrès 21], Bass-Burdzy flows
[Bass-Burdzy 99, Warren 05]
- ▶ [Aïdékon-Wang-Yu, 24+]:
skew Brownian flow [Burdzy-Chen, 01, Burdzy-Kaspi, 04]

Bifurcation points in BESQ flows

- ▶ $a \mapsto \tau_a^u = \inf\{t \geq 0: L_\xi(t, u) > a\}$ is càdlàg.
- ▶ $\tau_{a-}^u = \lim_{a' \uparrow a} \tau_{a'}^u = \inf\{t \geq 0: L_\xi(t, u) \geq a\}$
- ▶ fixed (a, u) : almost surely $\tau_{a-}^u = \tau_a^u$; **bifurcation point** (a, u) :

$(L_\xi(\tau_a^u, y) - L_\xi(\tau_{a-}^u, y), y \geq u)$ is a BESQ(0) excursion

- ▶ [Aidékon-Wang-Yu, 2024+] The set of bifurcation points has Hausdorff dimension $3/2$



Coupled squared Bessel flows

$\tau_a^r = \inf\{t \geq 0: L_\xi(t, r) > a\}$. Consider strong solutions of the SDE

$$\mathcal{R}_{(a,r)}(y) := a + 2 \int_r^y \mathbf{W}([0, L_\xi(\tau_a^r, s)], ds) + (\delta \mathbb{1}_{\{y \leq 0\}} + 0 \mathbb{1}_{\{y \geq 0\}})(y - r),$$

$$\mathcal{B}_{(a,r)}(y) := a + 2 \int_r^y \mathbf{W}([0, L_\xi(\tau_a^r, s)], ds) + (0 \mathbb{1}_{\{y \leq 0\}} - \delta \mathbb{1}_{\{y \geq 0\}})(y - r),$$

driven by **the same white noise $\mathbf{W}$** on $\mathbb{R}_+ \times \mathbb{R}$.

- ▶ $\mathcal{R}_{(a,r)}(y)$: BESQ($\delta \mid_0 0$) flows
- ▶ $\mathcal{B}_{(a,r)}(y)$: BESQ($0 \mid_0 -\delta$) flows
- ▶ With $W(g) = \int g(\rho_t) \text{sgn}(\xi_t) d\xi_t$, we have

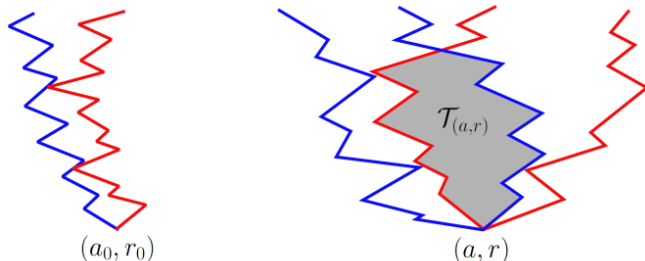
$$\mathcal{R}_{(a,r)}(y) = L_\xi(\tau_a^r, y)$$

Exceptional points and spindles

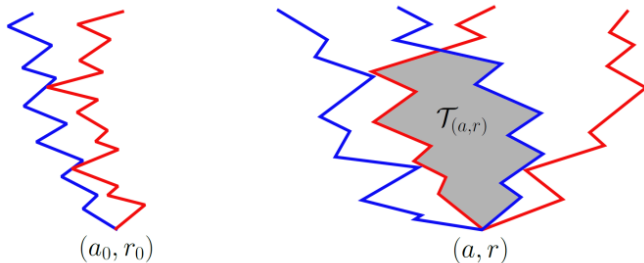
- ▶ **Comparison principle:** Almost surely, for all $(a, r) \in \mathbb{R}_+ \times \mathbb{R}$ and $y \geq r$, $\mathcal{B}_{(a,r)}(y) \leq \mathcal{R}_{(a,r)}(y)$.
- ▶ **double bifurcation points** $(a, r) \in \mathbb{R}_+ \times \mathbb{R}$ if

$$\mathcal{T}_{(a,r)} := \{(b, x) \in \mathbb{R}_+ \times (r, \infty) : \mathcal{B}_{(a,r)}(y) > b > \mathcal{R}_{(a-,r)}(y)\} \neq \emptyset$$

- ▶ [Aidékon-Wang-Yu 2024+]: the family of double bifurcation points is non-empty iff the parameter difference $\delta \in (0, 2)$;
Hausdorff dimension: $\min(2 - \delta, \frac{3-\delta}{2}) < 3/2$



A random partition of the half-plane



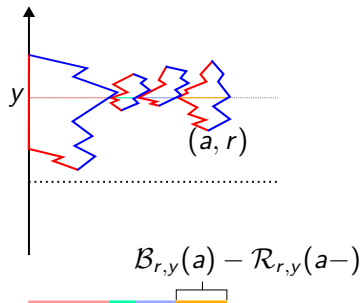
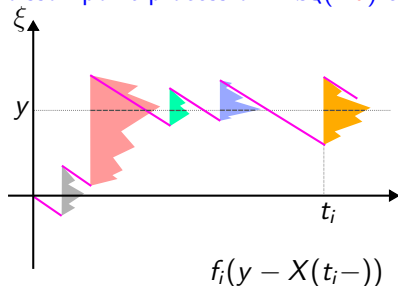
- ▶ Dichotomy: $\begin{cases} \text{either } \mathcal{T}_{(a,r)} \subseteq \mathcal{T}_{(a',r')} \\ \text{or } \mathcal{T}_{(a,r)} \cap \mathcal{T}_{(a',r')} = \emptyset \end{cases}$
- ▶ Define a partition $\mathcal{P} := \bigcup_{(a,r) \in \mathbb{R}_+ \times \mathbb{R}} \mathcal{T}_{(a,r)}$
- ▶ $(\mathbb{R}_+ \times \mathbb{R}) \setminus \mathcal{P}$ has zero Lebesgue measure

Main result [Aïdékon-S.-Wang 2025+]

We define a **change of time** η_t , such that

- ▶ $X_t := \xi(\eta_t)$ is a spectrally positive $(1 + \delta/2)$ -stable Lévy process;
- ▶ The range of $\rho(\eta_t)$: the gasket of the partition $(\mathbb{R}_+ \times \mathbb{R}) \setminus \mathcal{P}$
- ▶ **Bijection**: for any jump time $t \geq 0$ of X ,
 $\rho(t-) = (a, r)$ for some $\mathcal{T}_{(a,r)}$
- ▶ Width of $\mathcal{T}_{(\rho(t-))}$, $t \geq 0$:

Poisson point process of BESQ($-\delta$)-excursions



Stable shredded spheres

- ξ : the scaffolding $(1 + \delta/2)$ stable process

$$D^u(t_1, t_2) = \xi_{t_1} + \xi_{t_2} - 2 \inf_{t \in [t_1 \wedge t_2, t_1 \vee t_2]} \xi_t,$$

$$D^d(t_1, t_2) = 2 \sup_{t \in [t_1 \wedge t_2, t_1 \vee t_2]} \xi_t - \xi_{t_1} - \xi_{t_2}.$$

- Equivalence relation on $\mathbb{R}_+$:

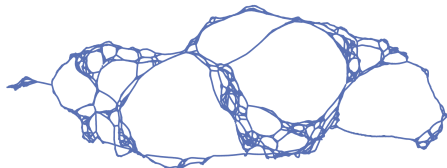
$$t_1 \sim_\xi t_2 \text{ iff } \min(D^u(t_1, t_2), D^d(t_1, t_2)) = 0$$

Theorem (Aidékon-S.-Wang 2025+)

The gasket $(\mathbb{R}_+ \times \mathbb{R}) \setminus \mathcal{P}$ is homeomorphic to $\mathbb{R}_+ / \sim_\xi$.

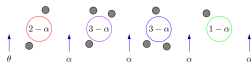
[Björnberg–Curien–Stefánsson, AOP2022]:

- ξ_e : **normalized excursion** above the infimum of ξ
- **Stable shredded sphere**: $[0, 1] / \sim_{\xi_e}$ for similarly defined $\sim_{\xi_e}$: scaling limits of certain random planar maps

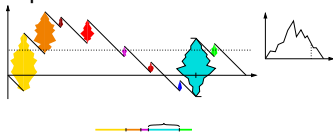


Summary

- Scaling limits of Chinese restaurant processes



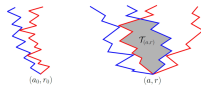
- Representation as a marked Lévy process



- BESQ flows from Brownian local time

$$L_{\xi}(\tau_a^u, y) = a + 2 \int_u^y W([0, L_{\xi}(\tau_a^u, s)], ds) + (0 \mathbb{1}_{\{y \geq 0\}} + \delta \mathbb{1}_{\{y \leq 0\}})(y - u),$$

- Partition of half-plane from **coupled** BESQ flows



- Connected to and stable shredded spheres





E. Aïdékon, Q. Shi and C. Wang.

Stochastic flows and marked stable processes.

[arXiv:2512.05466v1](#).



Q. Shi and M. Winkel.

Up-down ordered Chinese restaurant processes with two-sided immigration, diffusion limits and emigration.

[Annals of Applied Probability](#), to appear.

Thank You!